

Fire ALarm System

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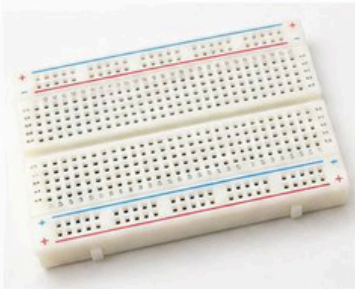
1. INTRODUCTION

This project demonstrates the design and implementation of a simple fire alarm circuit using a thermistor sensor. The circuit is designed to detect heat and fire using a $10k\Omega$ NTC thermistor. When the thermistor is exposed to heat, its resistance decreases and sends a signal to the transistor. The transistor works as an electronic switch that activates the buzzer alarm. The circuit is powered by a 9V battery. This type of system can be used in basic fire detection and safety applications.

2. EQUIPMENT AND MATERIALS

1. Breadboard

Used to assemble the circuit without soldering.



2. Thermistor (NTC)

Senses heat and changes resistance according to temperature.



3. 2N2222 NPN Transistor

Acts as an electronic switch.



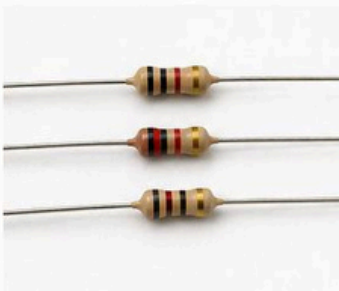
4. Buzzer

Produces sound when heat or fire is detected.



5. Resistors

Used to control voltage and current flow in the circuit.



6. Jumper Wires

Used for electrical connections between components.



7. 9V Battery

Provides power to the circuit.



8. 9V Battery Connector

Connects the battery to the breadboard securely.



3. CIRCUIT DIAGRAM

The following figure shows the complete fire alarm circuit using a thermistor sensor and transistor switching system.

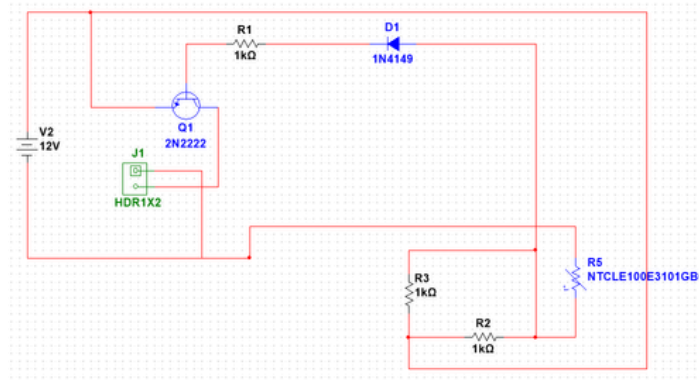


FIGURE (1): FIRE ALARM CIRCUIT DIAGRAM

4. CIRCUIT CONNECTIONS

The following figure shows the practical connections of the circuit components on the breadboard.

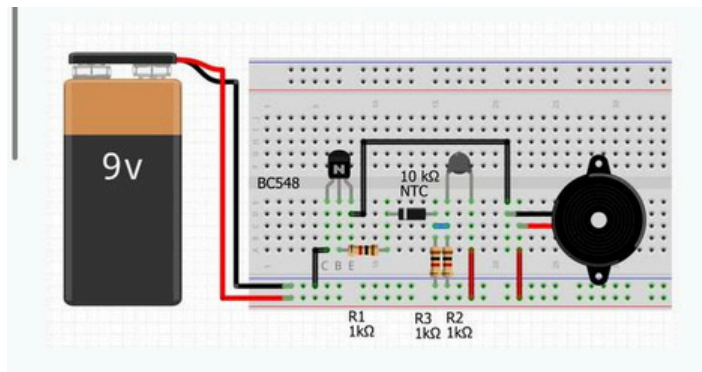


FIGURE (2): CIRCUIT CONNECTIONS ON BREADBOARD

5. PROJECT MODEL

The following figure shows the project model using a miniature house to represent a fire detection system inside a home environment.

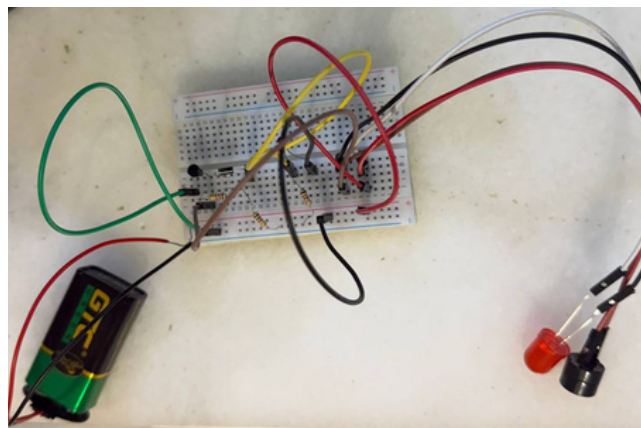


FIGURE (3): FIRE ALARM SYSTEM MODEL

6.WORKING PRINCIPLE

The circuit operates based on the resistance change of the thermistor (NTC). At normal temperature, the transistor remains off and the buzzer does not operate. When heat or fire is detected, the thermistor resistance decreases, causing the transistor to switch on and activate the buzzer alarm automatically.

7.CONCLUSION

This project successfully demonstrates a simple fire alarm circuit using a thermistor sensor. The alarm activates automatically when high temperature or fire is detected. The circuit is simple, low-cost, and suitable for basic fire safety applications.